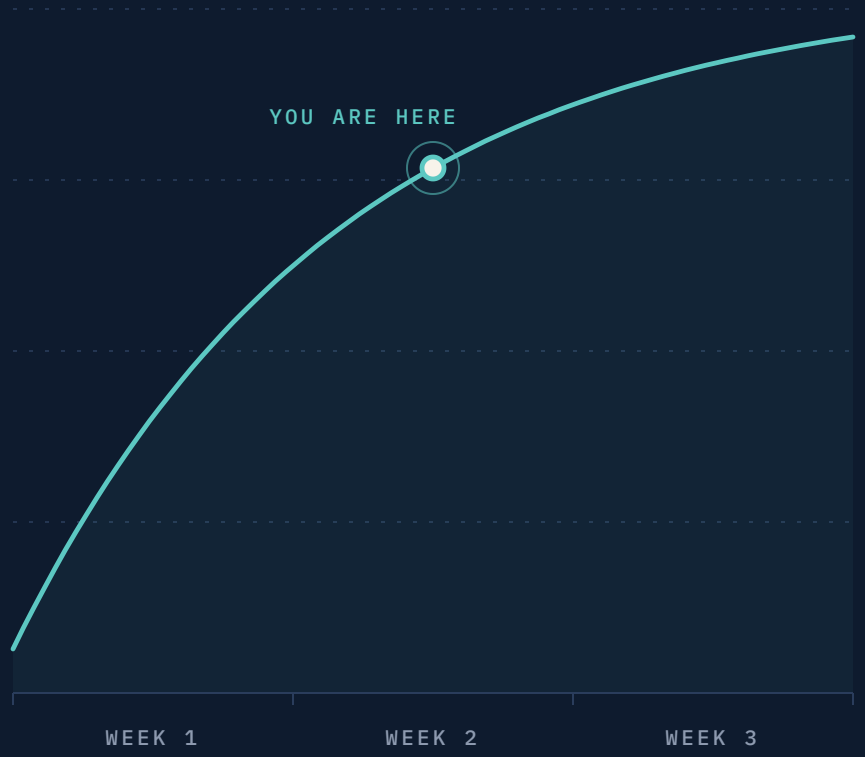
WEEK 2 · DAY 3 · OVERVIEW

ROC Curves and Choosing a Threshold



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ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 4 EXERCISES

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Problem Statement

BUSINESS CONTEXT

Day 7 ended with a model that flags a quarter of the defaults at 0.5, and a 0.5 that is ours to move.

Every threshold makes a different list of accounts to review, and nothing in the file says which list is right.

Today the desk sees all of those lists at once, scores the ranking behind them, and chooses one threshold by a rule written down before looking.

The file is Day 6's: 30,000 real card accounts in Taiwan in 2005, so nothing here carries over to another market or year.

SEX, **EDUCATION**, **MARRIAGE** and **AGE** are never features: a model using them would sort people by them, and leaving them out does not make a model neutral, because other columns can carry the same information.

A flag means "flagged at threshold t", never a credit decision about anyone.

OBJECTIVES

Draw the ROC curve by hand and with scikit-learn, score the whole curve with AUC and see what it reads, draw the precision-recall curve, choose a threshold on validation accounts by a stated rule, show why the accounts that choose cannot report, and read one calibration chart.

ABOUT 50 MIN WITH EXERCISES / 12 SECTIONS / 4 EXERCISES

Your Route Through Day 3

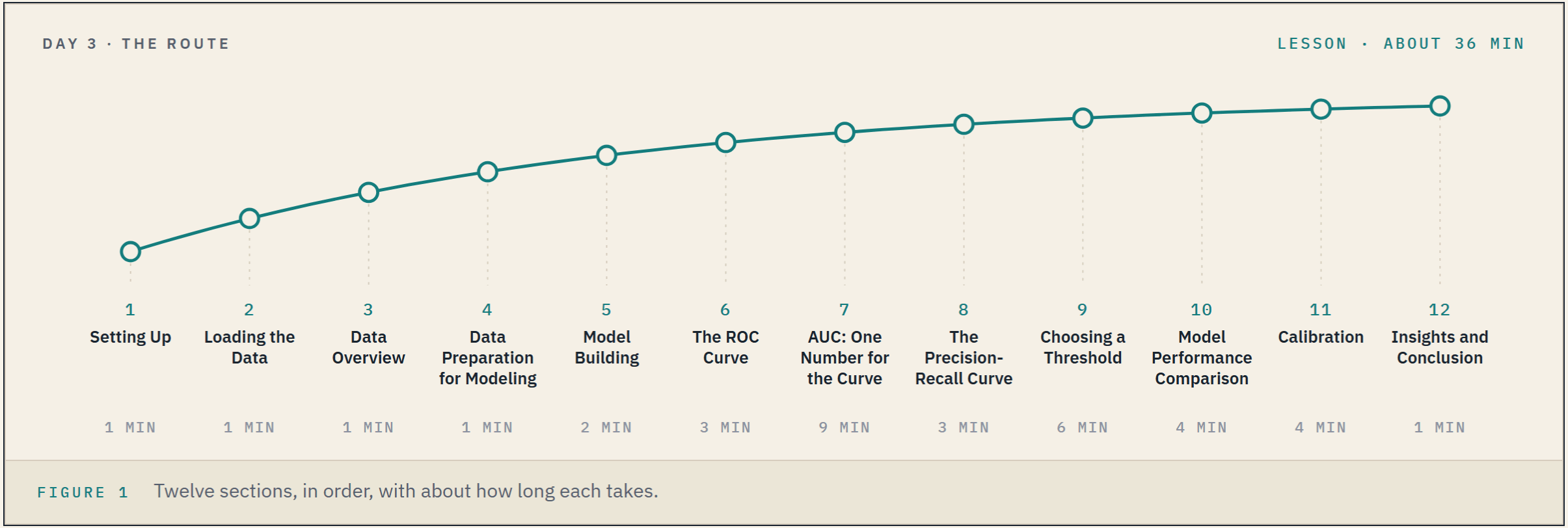


FIGURE 1 Twelve sections, in order, with about how long each takes.

Time. About 36 minutes for the lesson, then about 12 for the four exercises.

Fix the seed and load the libraries.

Setting Up

SEED AND DATA LOCATION

```
SEED = 42
DATA = "../../../data/"
if not Path(DATA + "benchmark.csv").exists():
    DATA = "https://raw.githubusercontent.com/lenamonj/python-fixed-income/main/data/"
```

THE CURVE TOOLS

```
from sklearn.metrics import accuracy_score, confusion_matrix, precision_score, recall_score, f1_score
from sklearn.metrics import roc_curve, roc_auc_score, precision_recall_curve, average_precision_score, auc
from sklearn.calibration import calibration_curve
```

ONE COLOUR FOR EACH KIND OF ACCOUNT

```
CELL_COLOURS = {"TN": "#56B4E9", "FP": "#DE8F05", "FN": "#D55E00", "TP": "#0173B2"}
```

Read the 30,000 card accounts.

Loading the Data

ONE ROW PER ACCOUNT

- **ID**: a row label, never a feature
- **LIMIT_BAL**: the credit limit, NT dollars
- **PAY_0**, **PAY_2** to **PAY_6**: repayment status, September back to April
- **BILL_AMT1** to **BILL_AMT6**: the bill, NT dollars
- **PAY_AMT1** to **PAY_AMT6**: the amount paid, NT dollars
- **SEX**, **EDUCATION**, **MARRIAGE**, **AGE**: about the person, never features
- **default payment next month**: 1 if the account defaulted the following month, 0 if not, **the target**

READ THE FILE

```
card = pd.read_csv(DATA + "credit_card_default.csv")
TARGET = "default payment next month"
```

Check the file's size and default rate.

Data Overview

SIZE AND DEFAULT RATE

```
print(card.shape)
print(card[TARGET].sum(), "accounts defaulted:", round(card[TARGET].mean() * 100, 2), "percent")
```

```
(30000, 25)
6636 accounts defaulted: 22.12 percent
```

WHAT THE FILE HOLDS

30,000 accounts and 25 columns; 6,636 defaulted, 22.12 percent.

The file is Day 6's and passed Day 6's checks, so we go straight to the split.

Cut the accounts into three sets.

Data Preparation for Modeling

THE FEATURES

```
PERSON_COLUMNS = ["SEX", "EDUCATION", "MARRIAGE", "AGE"]
FEATURES = [column for column in card.columns if column not in ["ID", TARGET] + PERSON_COLUMNS]
print(len(FEATURES), "features")
```

```
19 features
```

THE THREE SETS

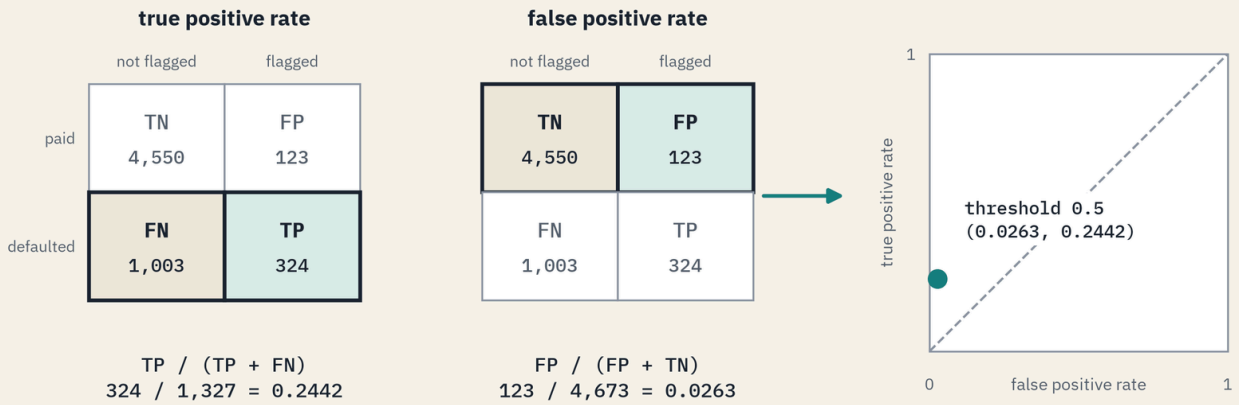
```
X_rest, X_test, y_rest, y_test = train_test_split(X, y, test_size=0.2, stratify=y, random_state=SEED)
X_train, X_valid, y_train, y_valid = train_test_split(X_rest, y_rest, test_size=0.25, stratify=y_rest, random_state=SEED)
print("Training:", len(X_train), " Validation:", len(X_valid), " Test:", len(X_test))
print("Defaults in validation:", y_valid.sum(), " Paid in validation:", (y_valid == 0).sum())
```

```
Training: 18000  Validation: 6000  Test: 6000
Defaults in validation: 1327  Paid in validation: 4673
```

Fit on training accounts, choose on validation accounts; the test accounts stay closed until Day 10.

Model Building

One threshold, two rates, one point: the matrix at 0.5



Each rate reads one row, so neither depends on how many accounts defaulted.

FIGURE 2 At 0.5 the true positive rate is 324 of 1,327 defaults and the false positive rate 123 of 4,673 paid accounts.

DISTINCT PROBABILITIES

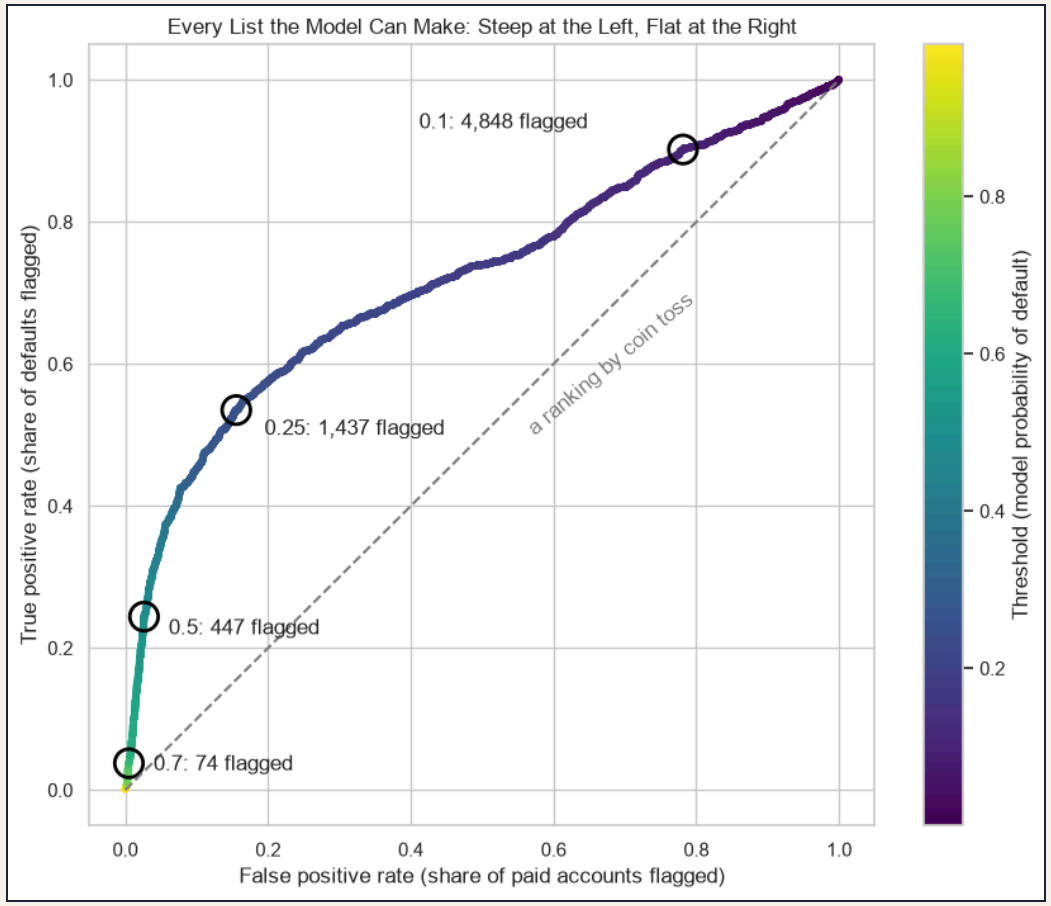
```
Accounts: 6000  Distinct probabilities: 5882
```

TWO RATES AT 0.5

```
True positive rate: 0.2442  False positive rate: 0.0263
```

One threshold gives one point on a chart of these two rates; every threshold gives a curve.

The ROC Curve



FOUR THRESHOLDS BY HAND

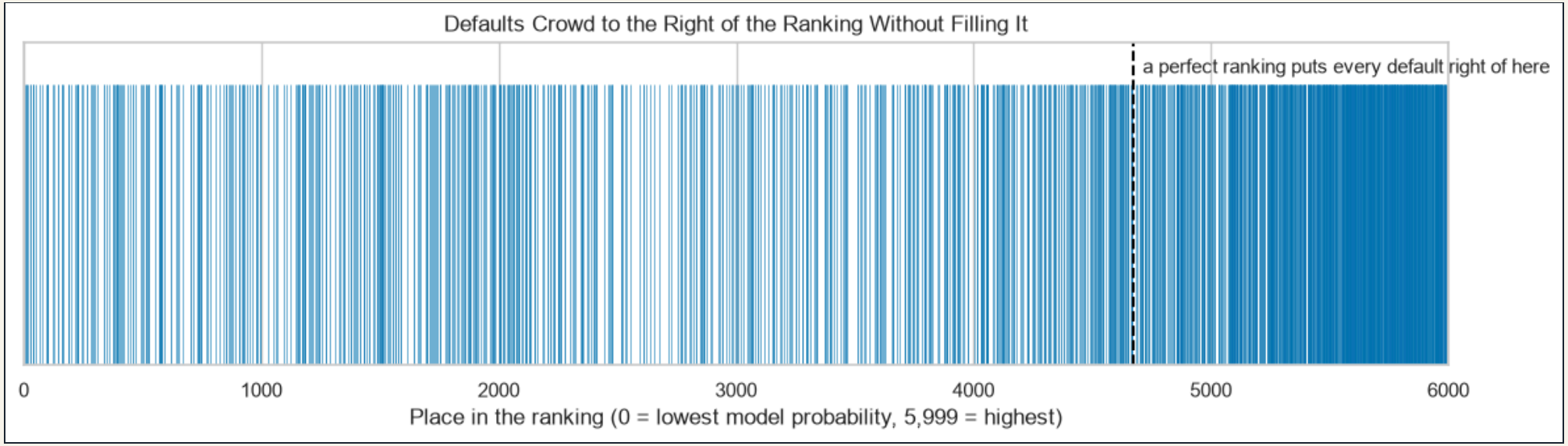
	THRESHOLD	FLAGGED	FPR	TPR
0	0.7000	74	0.0051	0.0377
1	0.5000	447	0.0263	0.2442
2	0.2500	1437	0.1556	0.5350
3	0.1000	4848	0.7813	0.9020

EVERY THRESHOLD

```
Thresholds: 5883  The first: inf
```

One curve holds every list; one number can score the whole curve.

AUC: One Number for the Curve



AUC FROM PROBABILITIES

```
Validation ROC AUC: 0.7190
Training ROC AUC:   0.7296
```

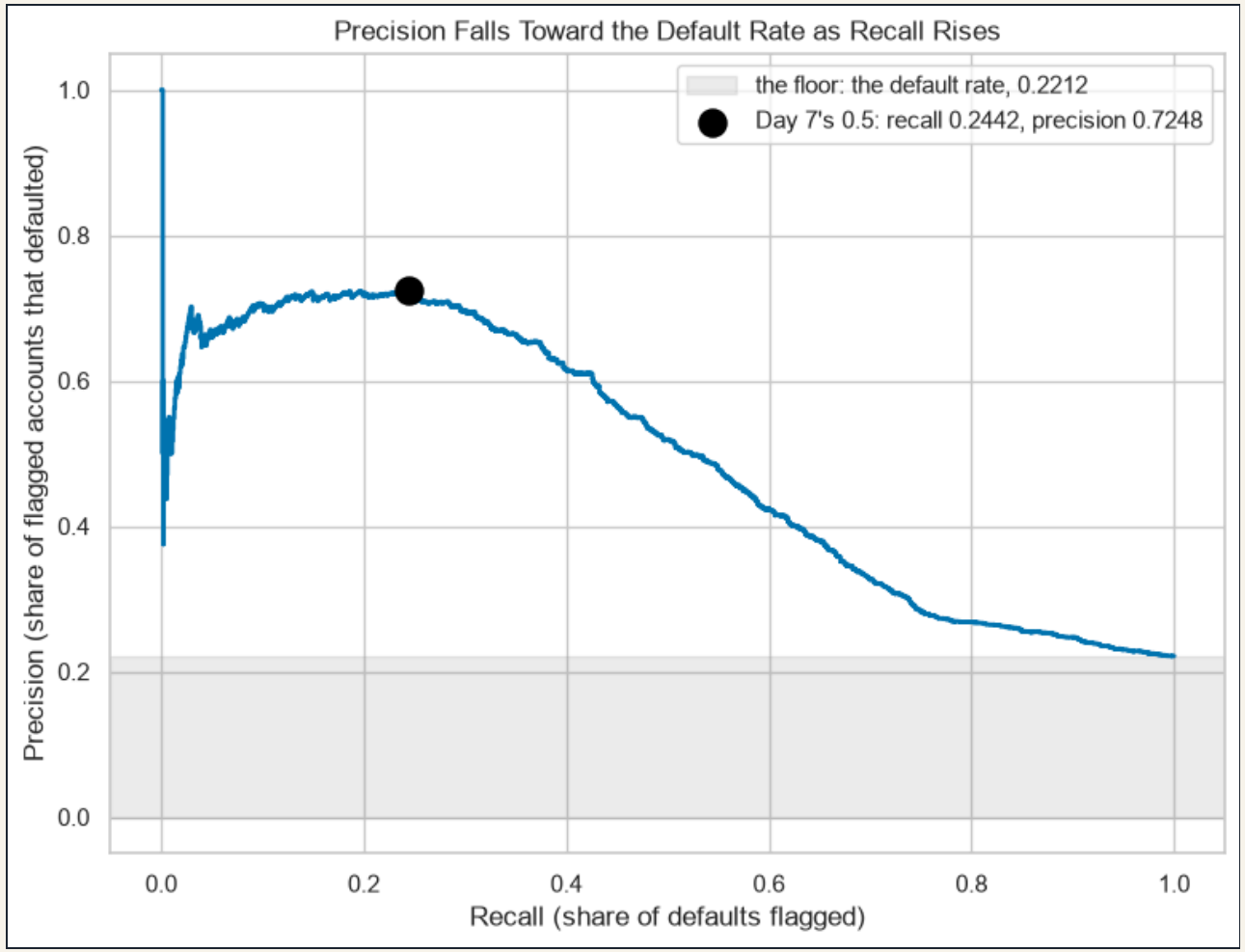
DEFAULTS IN EACH HALF

```
Defaults in the upper half: 949  in the lower half: 378
```

The area under the ROC curve, AUC, scores the whole curve in one number.

Trade precision against recall at every threshold.

The Precision-Recall Curve



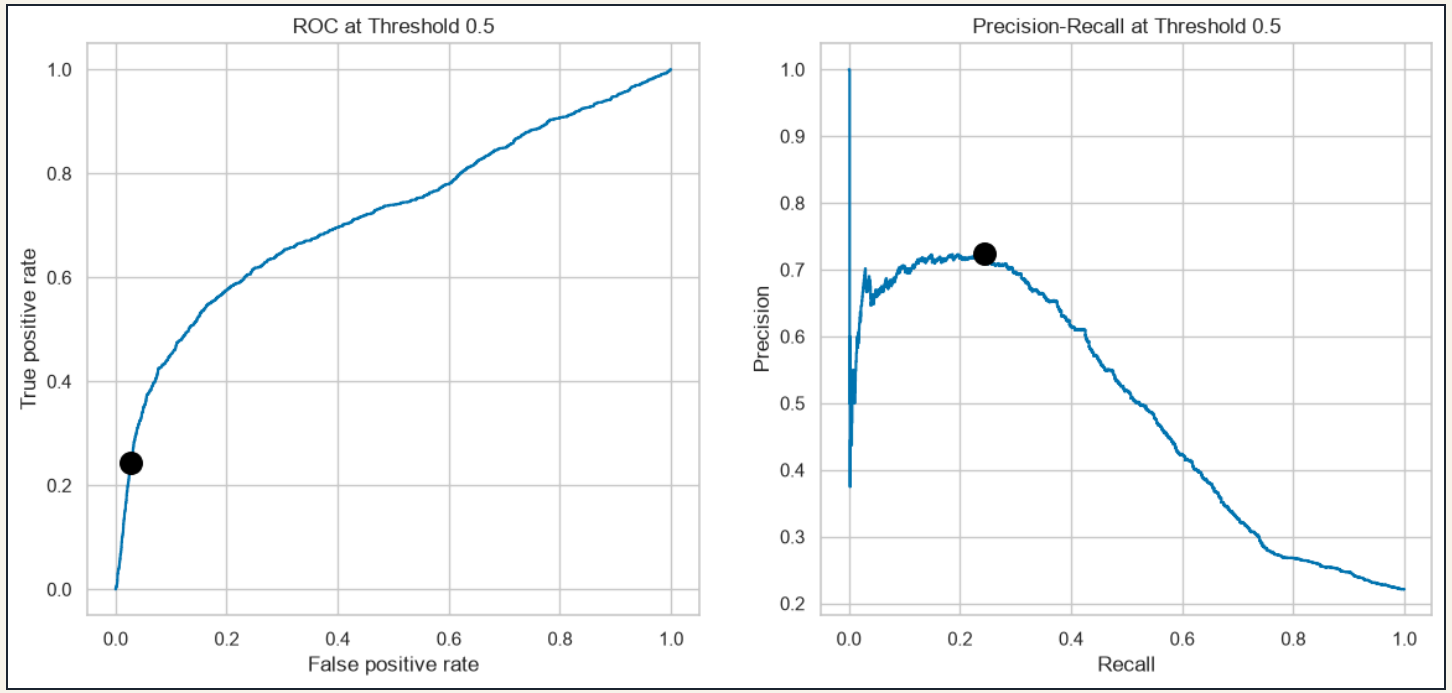
AVERAGE PRECISION AND THE TRAPEZOID

```
Average precision: 0.4946
Trapezoid area:    0.4940
```

BALANCED ROWS

```
Accounts: 2654  Default rate: 0.5
ROC AUC: 0.7158  Average precision: 0.7396
```

Choosing a Threshold



THE RULE: RECALL OF AT LEAST 0.70

```
Threshold: 0.2025
```

WHAT IT FLAGS

```
Flagged: 2843  Recall: 0.7001  Precision: 0.3268
Recall one step higher: 0.6993
```

About one flag in three is a default.

Score four thresholds, then test the choosing rows.

Model Performance Comparison

TRAINING ACCOUNTS

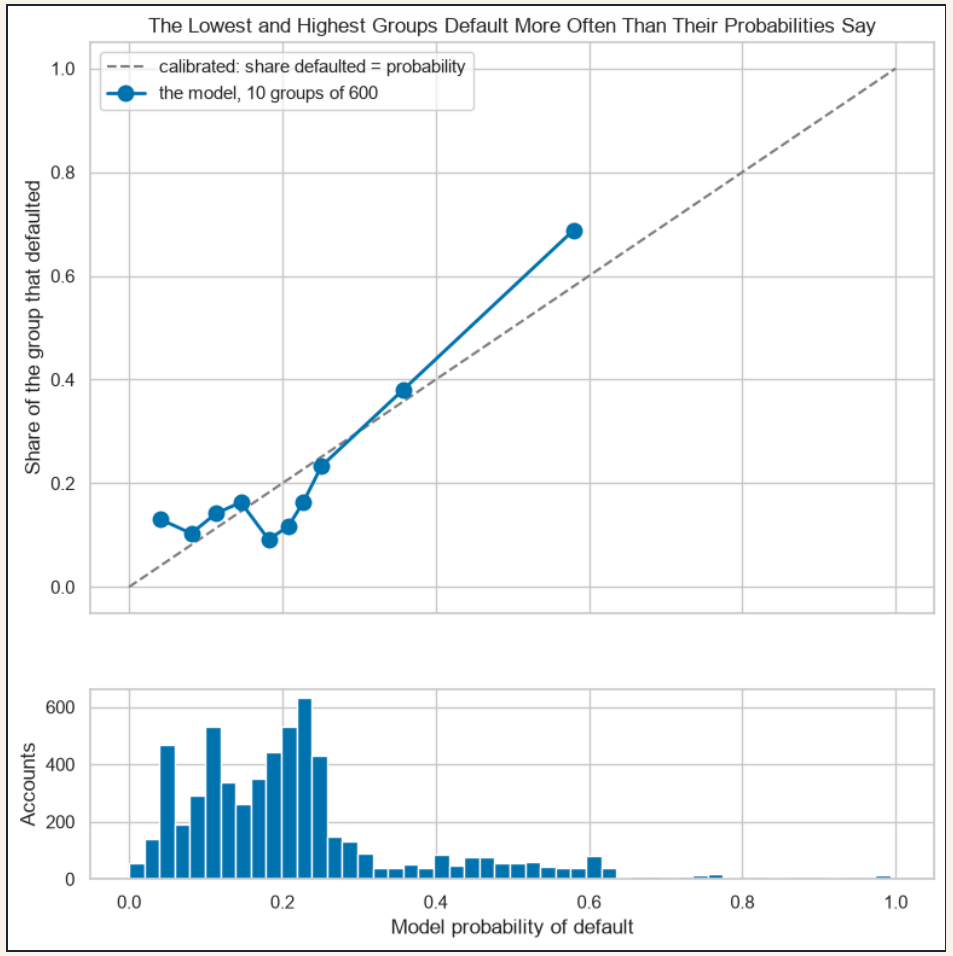
	0.5	RULE 0.70	YOUDEN
ROC AUC	0.7296	0.7296	0.7296
Accuracy	0.8131	0.6146	0.7736
Recall	0.2557	0.7089	0.5686
Precision	0.7179	0.3282	0.4898
F1	0.3770	0.4487	0.5263

VALIDATION ACCOUNTS

	0.5	RULE 0.70	YOUDEN
ROC AUC	0.7190	0.7190	0.7190
Accuracy	0.8123	0.6147	0.7707
Recall	0.2442	0.7001	0.5494
Precision	0.7248	0.3268	0.4837
F1	0.3653	0.4456	0.5145

We keep the rule, applied on validation accounts; on Day 10 the test accounts are opened once, to report.

Calibration

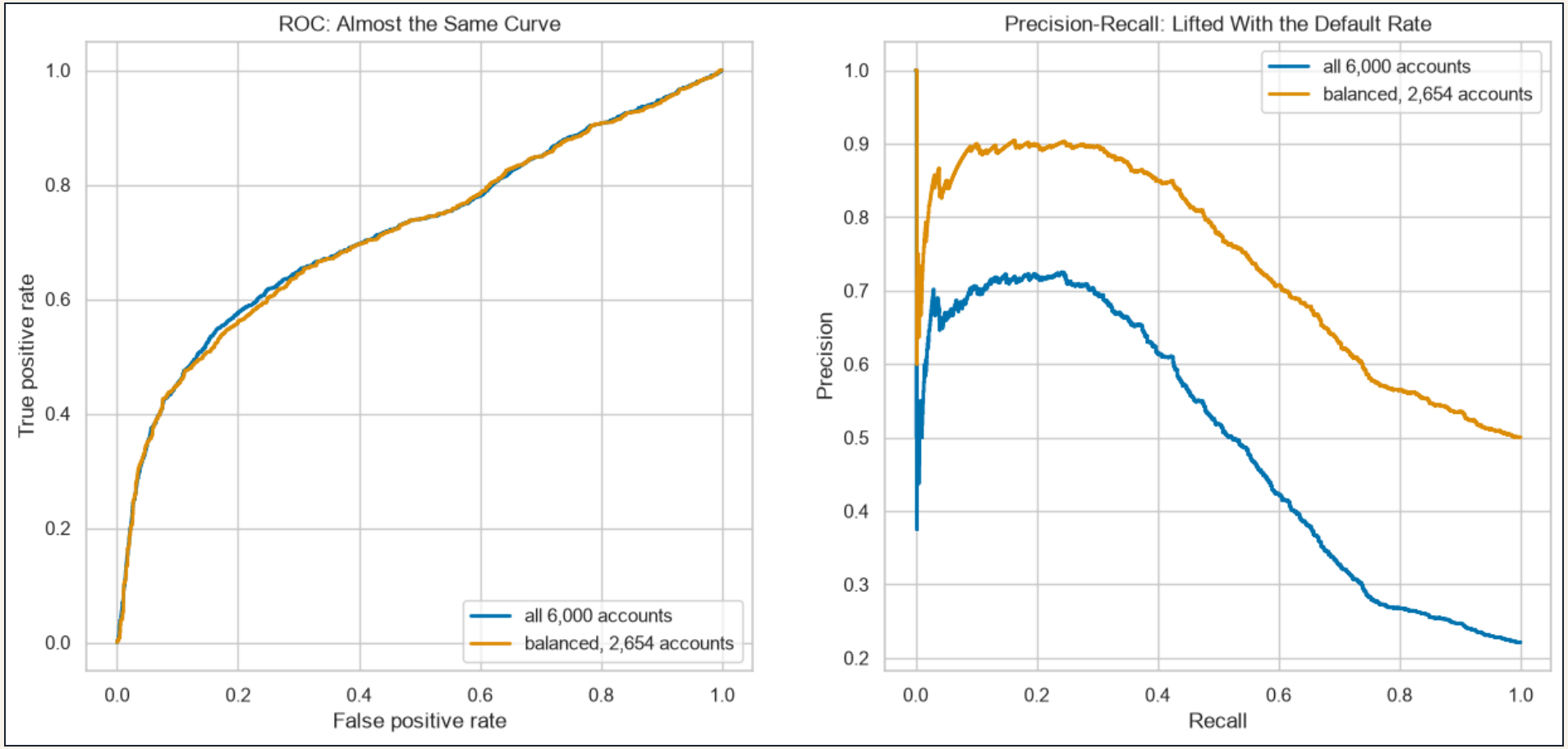


TEN GROUPS OF 600, LOWEST PROBABILITY FIRST

A model is **calibrated** when accounts given 0.3 default about three times in ten.

	MEAN PROBABILITY	SHARE DEFAULTED
0	0.0416	0.1300
1	0.0812	0.1033
2	0.1133	0.1417
3	0.1458	0.1633
4	0.1820	0.0917
5	0.2077	0.1167
6	0.2272	0.1633
7	0.2512	0.2333
8	0.3573	0.3800
9	0.5800	0.6883

Insights and Conclusion



Precision-recall moves with the default rate; ROC does not.

You Can Now

- Draw a ROC curve by hand and with `roc_curve`, and read AUC as a share of pairs.
- Draw a precision-recall curve and say why it moves with the default rate.
- Choose a threshold on validation accounts by a rule written down first, with Youden's J beside it.
- Show why the accounts that choose a threshold cannot report its score.
- Read a reliability diagram.

NEXT
WEEK 2 · DAY 4

Decision Trees

Day 9 meets a model of yes-or-no questions you can read: the decision tree.